Printed Pages: 02 **Sub Code: KEC-302**

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B. TECH
(SEM-III) THEORY EXAMINATION 2022-23
DIGITAL SYSTEM DESIGN

Time: 3 Hours ***Total Marks: 100***

Note: Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt *all* questions in brief. **2 x 10 = 20**

(a) Explain Minterm and Maxterm with example.
(b) Design and draw State Diagram for a 2 bit up/down counter
(c) Convert $(454.523)_{10}$ to an Hexadecimal number
(d) Draw a full adder using two half adders
(e) What do you mean by race around condition in JK Flip Flop?
(f) Describe figure of merit & noise immunity of TTL & CMOS ICs
(g) What is the basic concept of switched capacitors.
(h) Explain FAN-IN and FAN-OUT.
(i) What are the advantages and disadvantages of flash type ADC
(j) What is the difference between Multiplexer and Encoder

SECTION B

2. Attempt any *three* of the following: **10x3=30**

(a) What is magnitude comparator? Design a Single-bit comparator circuit using logic gates.
(b) Give the general procedure for converting a multilevel AND-OR diagram into an all NAND diagram. Implement the following Boolean function with NAND gates only.
$F(x, y, z) = \sum(1,2,3,4,5,7)$
(c) With neat diagram explain the operation of R-2R DAC
(d) Design a universal shift register that performs HOLD, SHIFT RIGHT, SHIFT LEFT, & LOAD operations.
(e) Draw and Explain a NAND gate in Totem Pole TTL Configuration.

SECTION C

3. Attempt any *one* part of the following: **10x1=10**

(a) Simplify the logic function using K-map
Y = ∑m (0, 2, 3, 4, 6, 7, 9, 11,16,18,19,20,22,23,25,27)
(b) Implement the function F = ∑m(0,1,3,4,7,8,9,11,14,15) using 8:1 mux.

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4. Attempt any *one* part of the following: 10x1=10

(a) Design a 3 bit up/down ripple counter

(b) Draw and explain the dual slope analog to digital convertor.

5. Attempt any *one* part of the following: 10x1=10

(a) Minimize y = ∑m(0,1,2,3,5,6,7,8,14,15) + d(4,11,13) using tabular method.

(b) Explain a Weighted Resistor digital to analog convertor.

6. Attempt any *one* part of the following: 10x1=10

(a) Explain Decoder with neat diagram. Implement the logic expression Y= Σm(2, 4, 6, 7) using decoder as ROM.

(b) Design a sequential circuit with two Flip Flops, A & B and one input x. When x=0, the State of the circuit remains the same when x=1 the circuit passes through the state transitions from 00 to 01 to 11 to 10 back to 00 & repeat.

7. Attempt any *one* part of the following: 10x1=10

(a) Draw CMOS inverter circuit and explain its working.

(b) Define PLD's. Implement the following function using PLA
F1 = ∑m(0,3,4,7)
F2 = ∑m(1,2,5,7)

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